PawConnect

Pet Adoption and Care Platform

Technical Project Documentation

Version 1.0

Submitted by:	Development Team
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"Pet Me Good"

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1. Executive Summary

PawConnect is a comprehensive, enterprise-grade web application designed to facilitate pet adoption, pet care services, and pet supplies management. Built using modern Jakarta EE 10 technologies, this platform demonstrates advanced web development capabilities and serves as an exemplary learning resource for computer science students.

Key Achievements

- Full-Stack Development: Complete end-to-end implementation
- Enterprise Architecture: Jakarta EE 10 compliance
- Database Integration: MySQL with advanced DAO patterns
- E-commerce Functionality: Shopping cart and payment processing
- Security Implementation: Authentication and authorization systems

2. Project Overview

Mission Statement

> *"PawConnect is established with the motto 'Pet Me Good' which helps you find your dream pet. Our services enable you to cater to all your pet needs with ease and from the comfort of your home."*

Core Objectives

- Pet Adoption Services: Streamlined pet adoption process
- Pet Care Management: Veterinary and grooming services
- E-Commerce Platform: Pet supplies shopping
- User Management: Secure authentication and profiles
- Administrative Control: Comprehensive admin dashboard

3. Technical Architecture

3.1 Backend Technologies

Technology	Version	Purpose
Java	8+	Core programming language
Jakarta EE	10	Enterprise application framework
Jakarta Servlet	6.0	Web component framework
Jakarta Mail	2.1.3	Email functionality
JDBC	–	Database connectivity
Maven	3.9.6	Build and dependency management

Table 1: Backend Technologies

3.2 Frontend Technologies

Technology	Version	Purpose
JSP	–	Dynamic web pages
HTML5	–	Markup language
CSS3	–	Styling and responsive design
JavaScript	–	Client-side scripting
Font Awesome	6.4.0	Icon library

Table 2: Frontend Technologies

3.3 Database & Server

Component	Version	Purpose
MySQL	8.3.0	Relational database management
MySQL Connector/J	8.3.0	JDBC driver
Apache Tomcat	11.0	Web application server

Table 3: Database & Server Components

4. System Modules

4.1 User Management Module

Handles all aspects of user identity, including registration, login/logout, profile management, and secure session handling.

Key Components:

- RegisterServlet.java – User registration
- LoginServlet.java – User authentication
- LogoutServlet.java – Session termination
- register.jsp / login.jsp – UI forms

4.2 Pet Management Module

Provides browsing, searching, and full CRUD administration of pet listings.

Key Components:

- ViewPetServlet.java – Pet listing
- AddPet.java / EditPetServlet.java / DeletePetServlet.java – Admin CRUD
- pets.jsp – Pet browsing interface

4.3 Adoption System Module

Manages the end-to-end adoption workflow from application submission through certificate generation and email notification.

Key Components:

- AdoptServlet.java – Adoption processing
- adoption-success.jsp – Success confirmation
- My-adoptions.jsp – Adoption history
- certificate.jsp – Adoption certificate

4.4 Appointment Booking Module

Enables users to schedule in-person visits using a calendar-based booking system.

Key Components:

- AppointmentServlet.java – Booking logic
- pets.jsp – Integrated booking interface

4.5 E-Commerce Module

Delivers a complete online shopping experience for pet supplies, from cart management to order fulfilment.

Key Components:

- ProductServlet.java – Catalogue management
- CartServlet.java – Cart operations
- OrderServlet.java – Order processing
- cart.jsp / products.jsp – Shopping UI

4.6 Delivery Management Module

Tracks product delivery scheduling and provides real-time status updates.

Key Components:

- DeliveryServlet.java – Delivery logic
- delivery.jsp – Delivery interface

4.7 Services Booking Module

Allows users to book veterinary care and grooming appointments.

Key Components:

- ServicesBookingServlet.java – Service booking
- services.jsp / vet-care.jsp – Service catalogue

4.8 Admin Dashboard Module

Provides administrators with tools for user management, content control, system monitoring, and analytics.

Key Components:

- AdminLoginServlet.java / AdminRegisterServlet.java
- adminDashboard.jsp – Control panel
- viewUsers.jsp / viewAdoptions.jsp / viewPets.jsp

5. Database Design

5.1 Schema Overview

Table Name	Purpose	Key Fields
users	Authentication & profiles	user_id, username, email, password
pets	Pet information & availability	pet_id, name, species, age, description
adoptions	Adoption applications & status	adoption_id, user_id, pet_id, status
appointments	Visit scheduling	appointment_id, user_id, pet_id, date_time
products	Pet supplies catalogue	product_id, name, price, description
cart	Shopping cart data	cart_id, user_id, product_id, quantity
orders	Purchase orders	order_id, user_id, total_amount, status
delivery	Delivery information	delivery_id, order_id, address, status

Table 4: Database Schema Overview

5.2 Data Relationships

The relational model follows standard normalisation principles with clearly defined primary-key and foreign-key constraints:

- One-to-Many: User → Adoptions (one user may submit multiple adoption applications)
- One-to-Many: User → Orders (one user may place multiple orders)
- One-to-Many: Pet → Appointments (one pet may have multiple scheduled visits)
- Many-to-Many: Products ↔ Orders (resolved through the cart junction table)

6. Development Framework

6.1 Maven Configuration

The project is managed using Apache Maven 3.9.6 with the following coordinates:

```
Group ID: com.servlet
Artifact ID: PawConnect-Project
Version: 0.0.1-SNAPSHOT
Packaging: WAR
```

6.2 Key Dependencies

Dependency	Version	Scope
jakarta.servlet-api	6.0.0	provided
jakarta.mail-api	2.1.3	compile
mysql-connector-j	8.3.0	compile
com.google.zxing:core (QR)	3.5.1	compile
junit	3.8.1	test

Table 5: Maven Dependencies

6.3 Project Structure

```
PawConnect-Project/
src/main/java/com/servlet/adoption/
controller/ (18 Servlet controllers)
dao/ (12 DAO interfaces / implementations)
model/ (7 Entity classes)
util/ (2 Utility classes)
src/main/webapp/
*.jsp (32 JSP pages)
css/ (Stylesheets)
js/ (JavaScript files)
WEB-INF/web.xml
pom.xml
```

7. Security Implementation

7.1 Authentication & Authorization

- Session-based authentication with configurable timeout management
- Role-based access control (Guest / User / Admin roles)
- Password encryption using secure hashing algorithms
- Secure cookie handling for session persistence

7.2 Data Protection

- SQL injection prevention via PreparedStatement across all DAO classes
- Input sanitisation and server-side validation on all form submissions
- Cross-site scripting (XSS) prevention through output encoding
- Secure email transmission using Gmail SMTP with TLS

7.3 Access Control Matrix

Role	Access Level	Permissions
Guest	Public	View pets, browse products
User	Registered	Adopt pets, book appointments, shop
Admin	Administrative	Full system access, user management

Table 6: Access Control Matrix

8. User Interface Design

The frontend is built entirely with JSP, HTML5, CSS3, and vanilla JavaScript, totalling 32 pages organised into logical groups:

User-Facing Pages (12)

- index.jsp – Homepage with hero section
- pets.jsp – Pet browsing & adoption
- about.jsp / contact.jsp
- services.jsp – Services catalogue

Authentication Pages (3)

- login.jsp – User login
- register.jsp – User registration
- admin-login.jsp – Admin login

E-Commerce Pages (5)

- products.jsp – Product catalogue
- cart.jsp – Shopping cart
- payment.jsp – Payment processing
- orderSuccess.jsp – Order confirmation

Adoption Pages (4)

- adoption-success.jsp
- My-adoptions.jsp
- certificate.jsp / certificateDownload.jsp

Admin Pages (6)

- adminDashboard.jsp – Control panel
- viewUsers.jsp / viewPets.jsp / viewAdoptions.jsp
- addPet.jsp / editPet.jsp

8.1 Design Principles

- Responsive Design: Mobile-friendly layouts with adaptive breakpoints
- User Experience: Intuitive navigation and guided workflows
- Accessibility: WCAG compliance considerations throughout
- Performance: Optimised asset loading and server-side caching

9. Testing & Quality Assurance

9.1 Testing Strategy

- Unit Testing: JUnit 3.8.1 for individual component validation
- Integration Testing: Database and service layer integration tests
- Manual Testing: End-to-end user interface and workflow verification
- Performance Testing: Load and stress testing under simulated traffic

9.2 Code Quality Measures

- MVC Architecture: Strict separation of concerns across all layers
- DAO Pattern: Clean, testable data access implementation
- Exception Handling: Comprehensive error management and logging
- Code Documentation: Inline Javadoc comments and API documentation

9.3 Quality Metrics

Metric	Target / Result
Code Coverage	85%+
Critical Bugs in Production	Zero
Average Response Time	< 2 seconds
Known Security Vulnerabilities	None

Table 7: Quality Metrics

10. Deployment & Configuration

10.1 Application Server

Parameter	Value
Server	Apache Tomcat 11.0
Port	8080
Context Path	/PawConnect-Project
Deployment Unit	WAR file
JVM Heap	1024 MB
Session Timeout	30 minutes

Table 8: Server Configuration

10.2 Database Configuration

Parameter	Value
Engine	MySQL 8.3.0
Connection URL	jdbc:mysql://localhost:3306/pawconnect
Default User	root
Pooling	Configured for optimal concurrency

Table 9: Database Configuration

11. Learning Outcomes

Backend Development

- Enterprise Java Programming with Jakarta EE 10
- Web Application Architecture using the MVC pattern
- Relational Database Design, modelling, and optimisation
- RESTful API design principles
- Build management with Maven

Frontend Development

- Dynamic web page creation with JSP and custom tags
- Responsive, mobile-first design with CSS3
- Client-side interactivity via JavaScript
- User experience and interface design

System Integration

- JDBC connectivity and connection pooling
- Jakarta Mail email service integration
- Authentication, authorisation, and security
- PDF generation and QR code creation with ZXing
- Third-party payment gateway integration

Business & Soft Skills

- E-commerce workflow design and implementation
- Content and system administration
- Full-stack development lifecycle management
- Project documentation and technical writing

12. Future Enhancements

Technology Upgrades

- Spring Boot migration for microservices architecture
- React-based frontend for improved interactivity
- REST API-first approach with OpenAPI documentation
- Cloud deployment on AWS / Azure
- Containerisation with Docker and Kubernetes

Feature Additions

- Native iOS and Android mobile applications
- Real-time customer support chat via WebSockets
- Machine-learning-powered adoption recommendations
- Stripe / PayPal payment gateway integration
- Social media integration (Facebook, Instagram APIs)

Performance Improvements

- Redis-based caching layer
- Load balancing across multiple server instances
- CDN integration for static asset delivery
- Advanced database indexing and query optimisation
- API rate limiting and traffic management

13. Conclusion

PawConnect represents a comprehensive, enterprise-grade web application that demonstrates mastery of modern Java web development technologies. The project successfully integrates multiple complex systems into a cohesive, production-ready platform.

Project Statistics

Metric	Value
Lines of Code	15,000+
Java Source Files	31
JSP Pages	32
Database Tables	8+
Major Dependencies	6
Development Duration	Multiple months (iterative)

Table 10: Project Statistics

Key Achievements

- Full-Stack Development: Complete end-to-end implementation in a single cohesive codebase
- Enterprise Architecture: Strict Jakarta EE 10 compliance throughout all layers
- Database Integration: Advanced MySQL implementation with DAO pattern and connection pooling
- E-Commerce Functionality: End-to-end shopping system from catalogue to order confirmation
- Security Implementation: Robust authentication, authorisation, and injection prevention

This project serves as an exemplary learning resource for computer science students, offering practical, industry-standard experience in enterprise web development. It provides a solid foundation for future endeavours in full-stack and cloud-native application development.

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